HAMED ADAM

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Profile

A driven, ambitious and diligent Theoretical Physics student with both research and engineering experience. Co-founded Lancaster University's Rocketry Society and worked as an experimental particle physics intern on LArTPC neutrino detection for Fermilab's DUNE experiment. Placed 2nd in the Royal Statistical Society hackathon while holding six part-time roles, from lab assistant to student recruitment, alongside full-time study. Eager to work in industrial R&D.

Education

Lancaster University **Lancaster, UK**
BSc Theoretical Physics | Predicted: Upper Second Class Honours *Oct 2024 – Jun 2027*
- Relevant modules: Computer Modelling, Complex Analysis, Vector Calculus, Linear Algebra

Manarat Jeddah International School **Jeddah, KSA**
A Levels and IGCSEs *Sep 2012 – Jun 2024*
- A Levels: Mathematics, Physics, Computer Science
- IGCSE: grades 9–7, including Mathematics

Experience

Particle Physics Group, Lancaster University **Lancaster, UK**
Experimental Particle Physics Intern *Aug 2026 – Oct 2026*
- Investigated neutrino interactions producing $\Lambda + K^+$ in the SBND liquid argon time projection chamber.
- Built a Python analysis pipeline on Pandora-reconstructed simulated events, using XGBoost boosted decision trees (BDTs) and a convolutional neural network (CNN) for particle identification and signal selection.
- Wrote a 20-page report on machine learning in particle physics and on optimising ML models for particle reconstruction.

Nonlinear and Biomedical Physics Group, Lancaster University **Lancaster, UK**
Research Intern *Jun 2026 – Aug 2026*
- Researched chronotaxic systems: nonautonomous oscillators with time-varying frequency whose amplitude *and* phase are both stable, giving them a resistance to phase perturbations absent in classical self-sustained oscillators.
- Contributed to the analysis of a 300-day Li-ion battery ageing experiment, using wavelet-based phase coherence (MODA, MATLAB) on impedance spectra reconstructed from current–voltage time series.
- Developed the case for a nonlinear, nonautonomous capacitor model of the voltage–current relationship, extending the fractional constant-phase-element description used in current models.

Lancaster University School of Physics and Astronomy **Lancaster, UK**
Lab Assistant *May 2025 – Present*
- Designed and set up undergraduate laboratory experiments on standing waves, ultraviolet radiation and X-rays.
- Diagnosed and fixed faults in oscilloscopes, signal generators and other laboratory equipment during practical sessions.
- Supported over 200 students from several sixth forms in producing research posters for national science competitions.

Lancaster Rocketry Society **Lancaster, UK**
Co-Founder and President *May 2026 – Present*
- Founded a 150+ member society to compete in the UKSEDS National Rocketry Championship and Satellite Design Competition against universities from across the UK.
- Led the design of competition hardware and engineered an avionics bay for dual-deployment recovery.
- Secured departmental and industry funding, along with exclusive lab access for the society.

Additional Experience

Picky Eater **Lancaster, UK**
Project Director *Oct 2025 – Present*
- Led technical development of a food discovery and ordering platform for independent restaurants.
- Managed an 8-developer team; designed the system architecture and MVP using Django and React.
- Defined sprint cycles and established coding standards across the team.

Lancaster University **Lancaster, UK**
External Relations, Widening Participation and Role Model Ambassador *Oct 2025 – Present*
- Coordinated open days, leading campus tours and delivering presentations to prospective students and their families.
- Ran sessions in local schools to encourage pupils from under-represented backgrounds to consider higher education.
- Represented the university at major outreach events with over 200 attendees.

Skills

Programming: Python, C/C++, MATLAB, Java, JavaScript/TypeScript (Node.js), SQL, Bash
Embedded & Systems: Assembly (x86), STM32 microcontrollers, PID control, real-time systems, PlatformIO, cross-compilation, hardware-in-the-loop testing, telemetry logging
Engineering & Design: SolidWorks, OpenRocket, CAD, FEA, technical drawing (GD&T), 3D printing/rapid prototyping, PCB & schematic design (KiCad)